

LLM Reasoning



Reasoning

- Tasks that require multiple steps of reasoning to solve

Arithmetic Reasoning (AR)

Question: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the

Answer: The answer is **5**.

Symbolic Reasoning (SR)

Question: Take the last letters of the words in "Elon Musk" and concatenate them

Answer: The answer is **nk**.

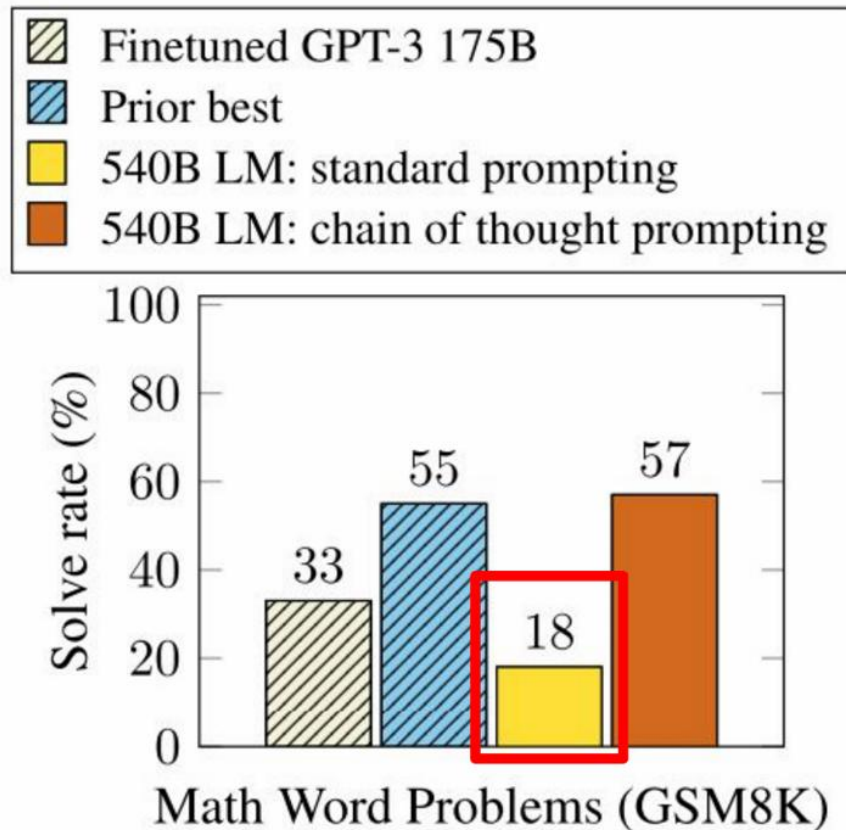
Commonsense Reasoning (CR)

Question: What home entertainment equipment requires cable? Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

Answer: The answer is **(c)**.

Reasoning Problems

GSM8K (arithmetic):



Proposed solution:

Chain of Thought (CoT) prompting

Few-shot standard prompting with even larger model (PaLM 540B) also does not work well.

Chain of Thought (CoT)

Few-Shot CoT

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Zero-Shot CoT

Large Language Models are Zero-Shot Reasoners

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Both papers
will appear in
NeurIPS'22!

Chain of Thought (CoT)

Definition:

A chain of thought is a **series of intermediate natural language reasoning steps** that lead to the final output.

⟨input, output⟩ demonstrations are replaced with ⟨input, **chain of thought**, output⟩

- Compositionality of the languages/problems
- **Benefits?**

Chain of Thought (CoT)

(Wei et al., 2022)

(a) Few-shot

Question: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: The answer is 11.

Question: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

Answer:

(Output) The answer is 8. ❌

(b) Few-shot-CoT

Question: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Question: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

Answer:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✅

Step-by-step Answer

Chain of Thought (CoT)

Zero-shot

Question: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

Answer: The answer (arabic numerals) is

(Output) 8 ✗

Zero-shot-CoT

Question: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

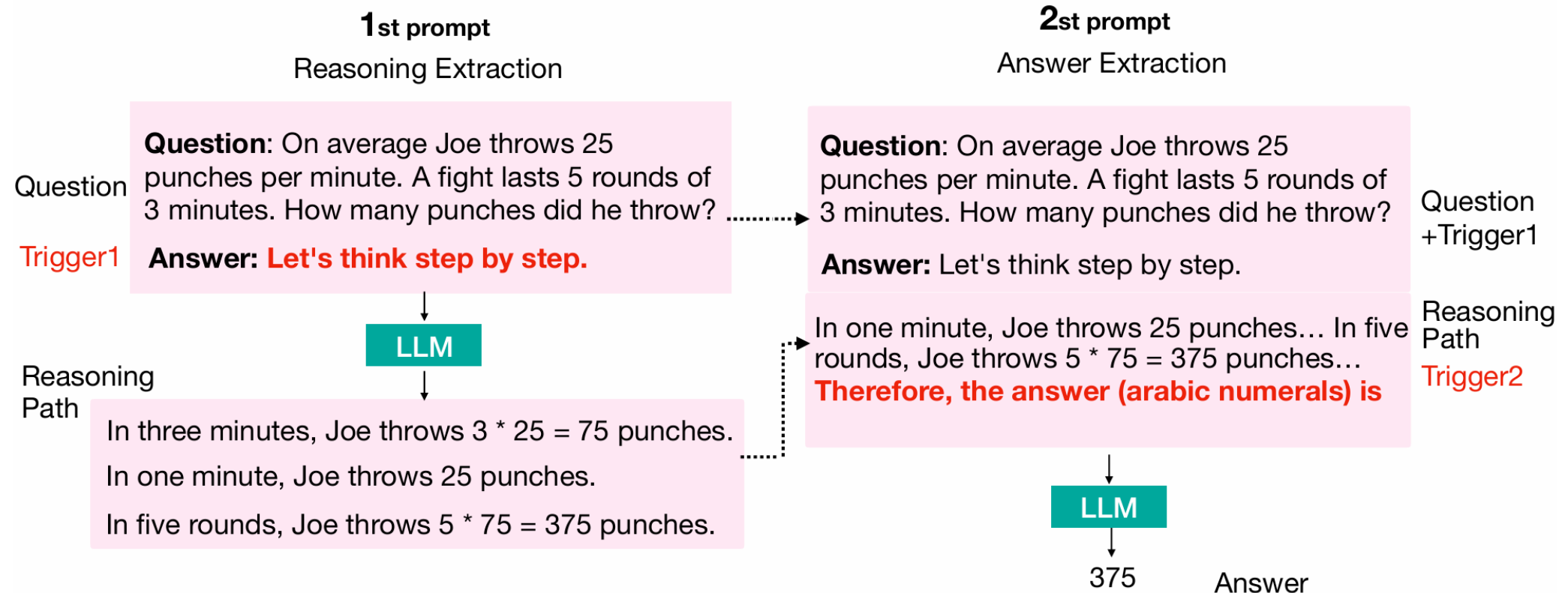
Answer: Let's think step by step.

(Output) (Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Two-stage Prompting
Step-by-step Answer

Zero-Shot Chain of Thought (CoT)

For zero-shot CoT, a **two-stage** prompting is applied:



Self-Consistency

Change greedy decode (single-path) to self-consistency (multi-path) in few-shot CoT:

Prompt with example chains of thought

Q: Shawn has five toys. He gets two more each from his mom and dad. How many toys does he have now?

A: Shawn started with 5 toys. 2 toys each from his mom and dad is 4 more toys. The final answer is $5+4=9$. The answer is 9.

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language model

Sample decode with diverse reasoning paths

She has $16 - 3 - 4 = 9$ eggs left. So she makes $\$2 * 9 = \18 per day. **The answer is \$18.**

This means she uses $3 + 4 = 7$ eggs every day. So in total she sells $7 * \$2 = \14 per day. **The answer is \$14.**

She eats 3 for breakfast, so she has $16 - 3 = 13$ left. Then she bakes muffins, so she has $13 - 4 = 9$ eggs left. So she has $9 * \$2 = \18 . **The answer is \$18.**

Majority vote

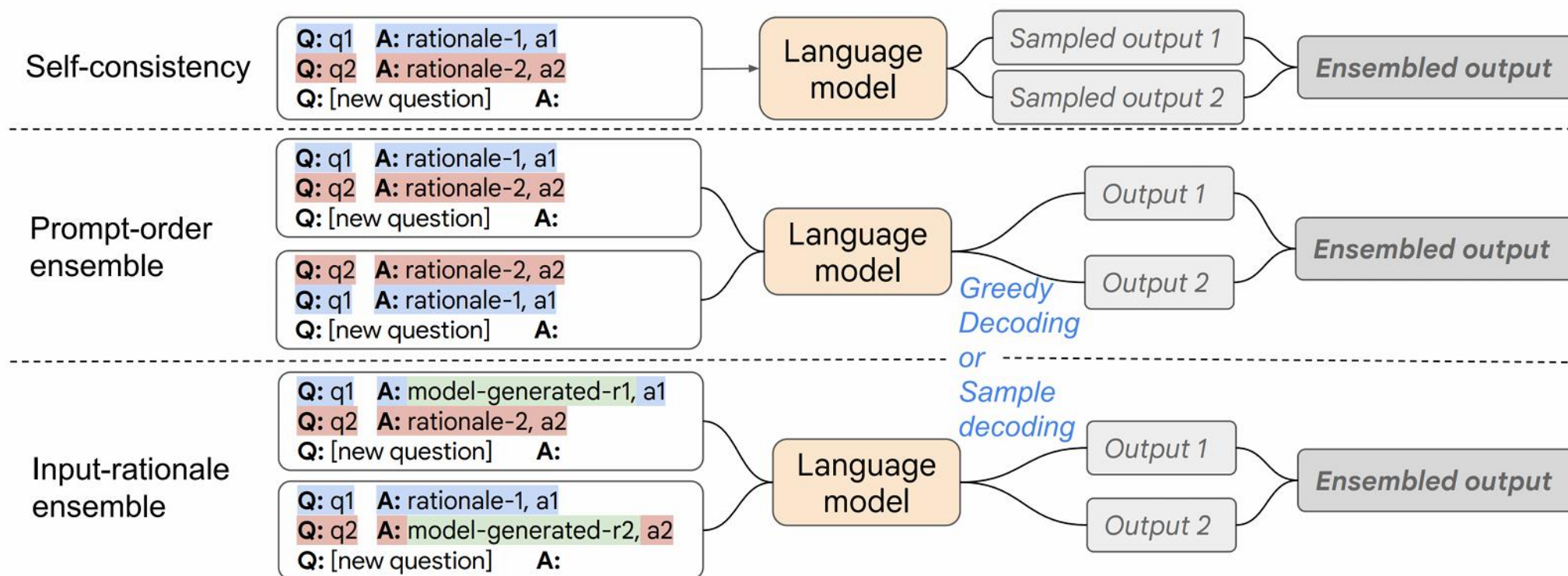
The answer is \$18.

Self-Consistency

	Method	GSM8K	CommonsenseQA
	Previous SoTA	$35^e / 57^g$	91.2^a
LaMDA (137B)	Greedy decode (Single-path)	17.1	57.9
	Self-Consistency (Multi-path)	27.7 (+10.6)	63.1 (+5.2)
PaLM (540B)	Greedy decode (Single-path)	56.5	79.0
	Self-Consistency (Multi-path)	74.4 (+17.9)	80.7 (+1.7)

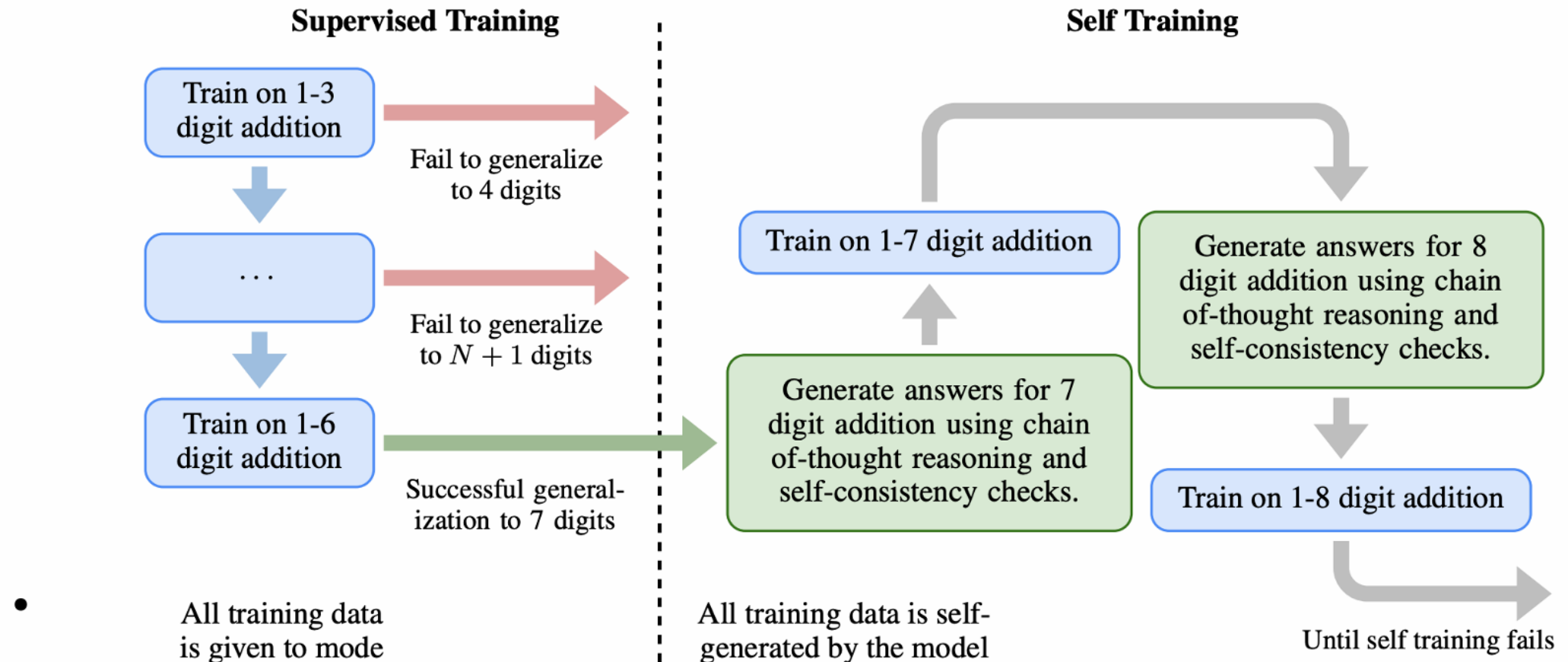
Ensembles in the input space

Use model-generated rationale in few-shot CoT:



Self-Education (SECToR)

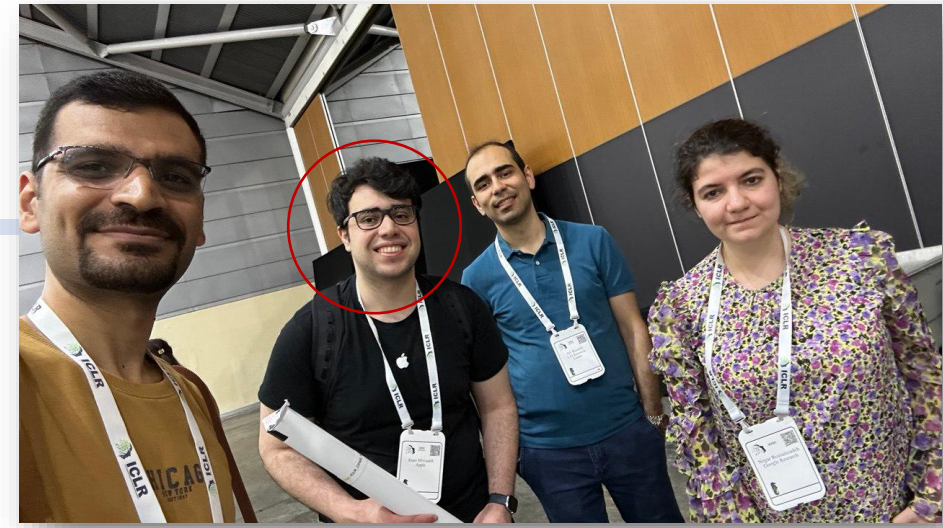
- Self-Education: LLMs can teach themselves new skills using CoT



Self-Education (SECToR)

- A supervised fine-tuning that includes training with curriculum learning is required.
- Self-education process:
 - Use CoT to generate solutions to problems that it could not otherwise solve
 - Then, the model is fine-tuned to generate these exact solutions without using CoT
 - This obtained model can now directly solve problems without using CoT
 - The self-learning process can continue accordingly

Iman's work



The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity

Parshin Shojae^{*†}
Maxwell Horton

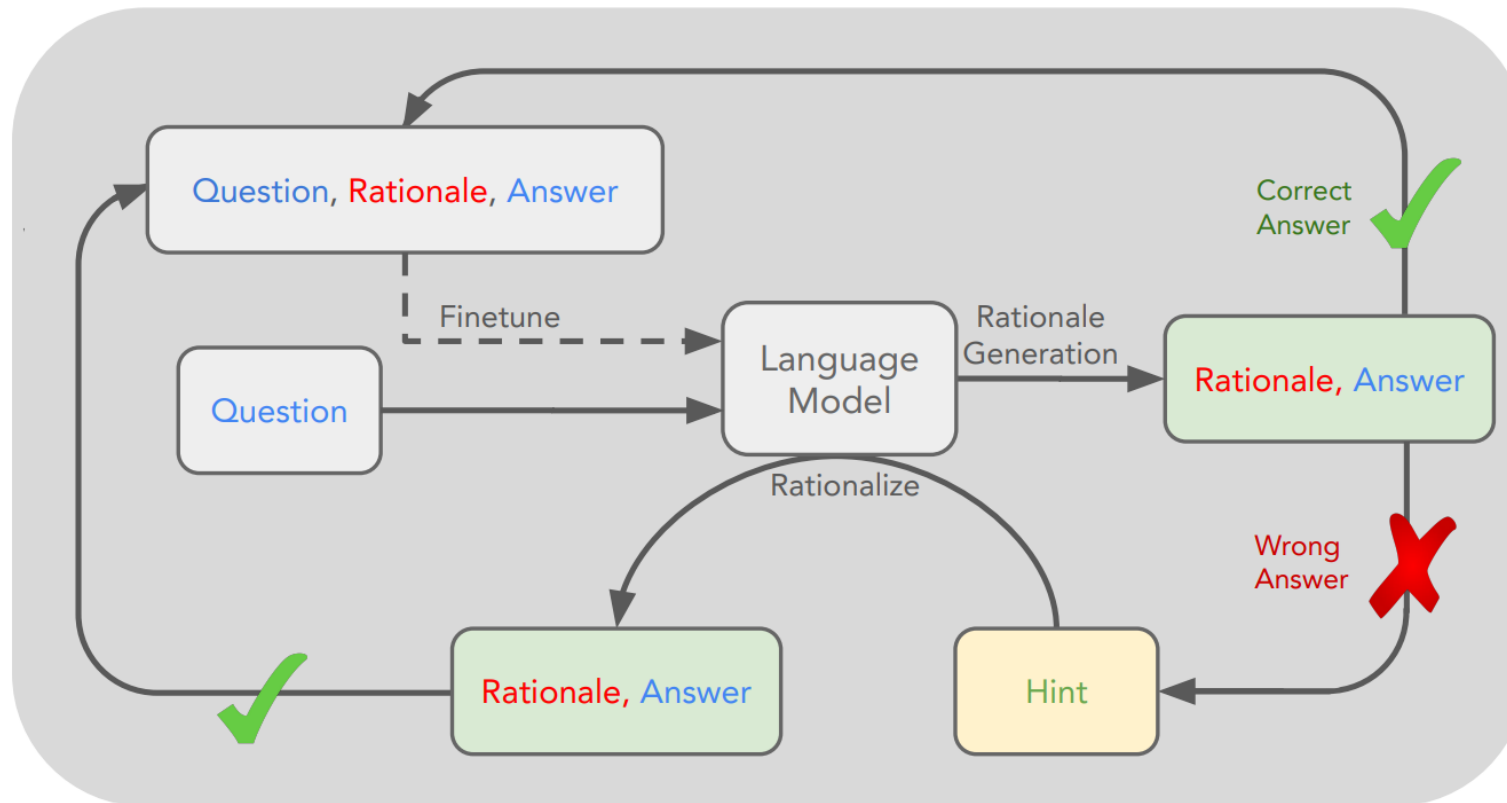
Iman Mirzadeh^{*}
Samy Bengio

Keivan Alizadeh
Mehrdad Farajtabar

Apple

STaR

Self-Taught Reasoner Bootstrapping Reasoning With Reasoning



Q: What can be used
to carry a small dog?

Answer Choices:

- (a) swimming pool
- (b) basket
- (c) dog show
- (d) backyard
- (e) own home

A: The answer must be
something that can be
used to carry a small
dog. Baskets are
designed to hold things.
Therefore, the answer
is basket (b).

V-STaR

Published as a conference paper at COLM 2024

V-STaR: Training Verifiers for Self-Taught Reasoners

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Aaron Courville  Alessandro Sordoni   Rishabh Agarwal 

 Mila, Université de Montréal

 Microsoft Research

 University of Edinburgh

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V-STaR

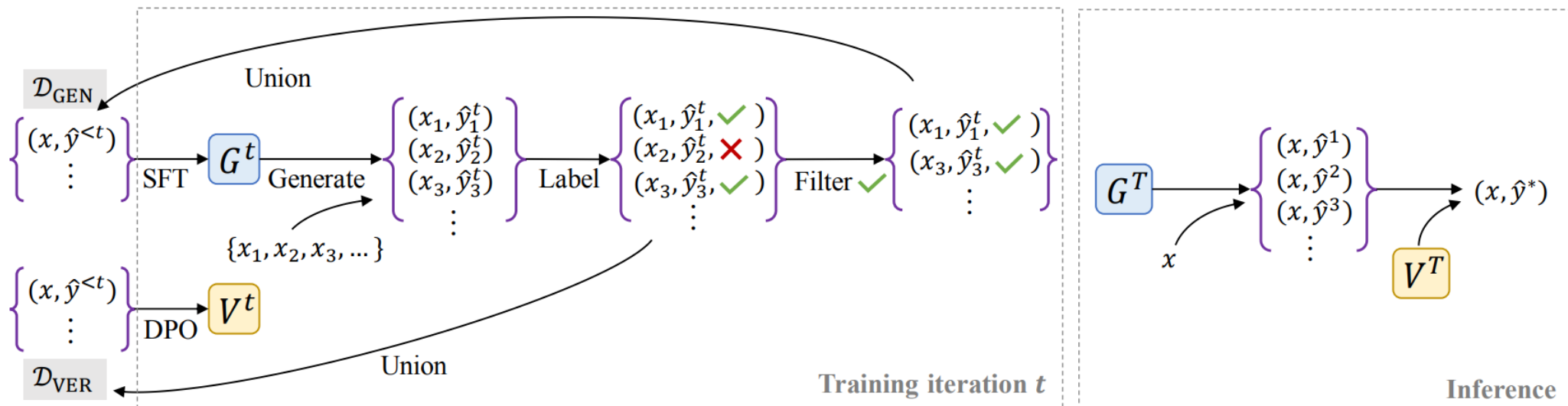
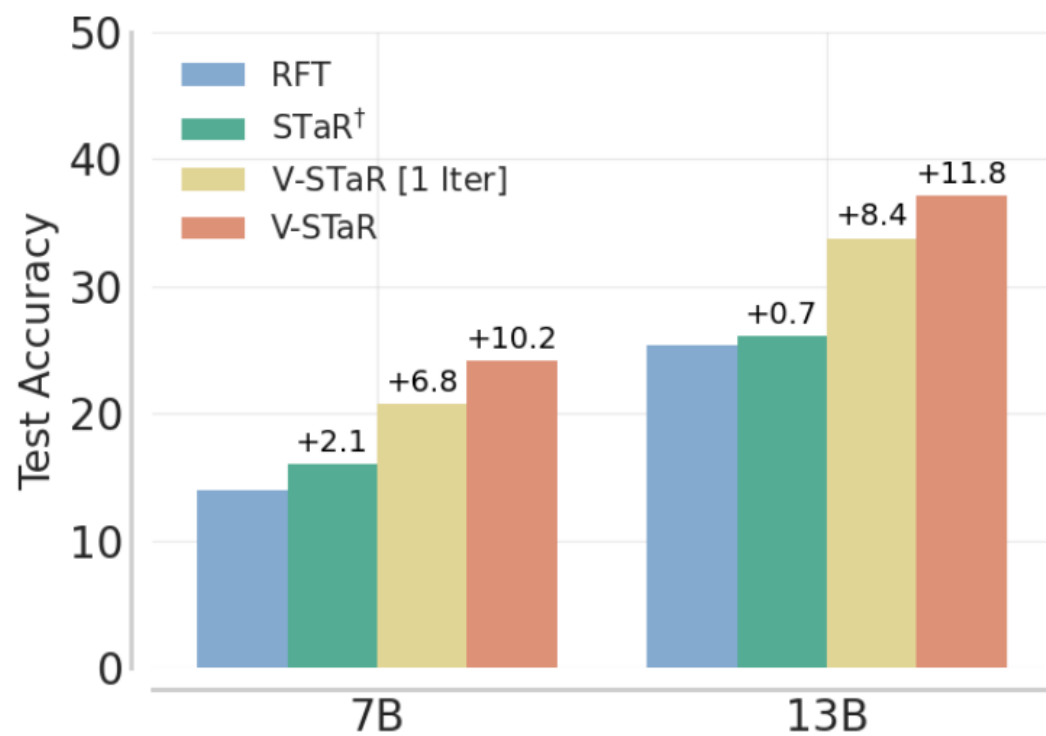
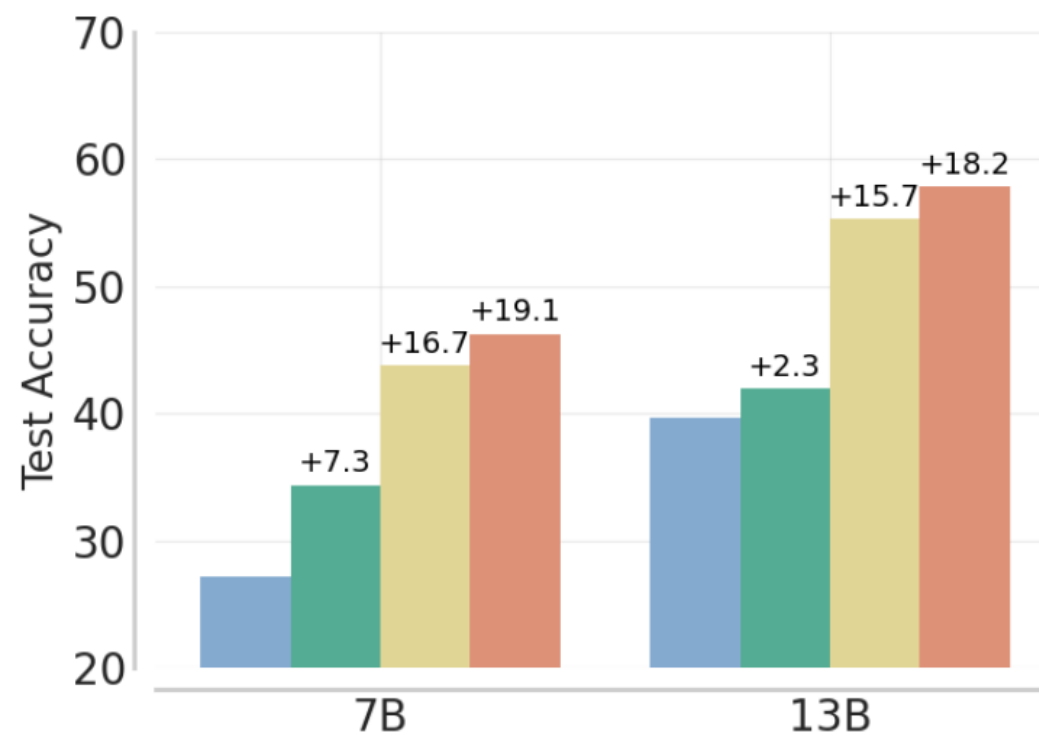


Figure 1: **Generator and verifier training in V-STaR.** **Left:** In each training iteration, the generator G^t is fine-tuned (from a pretrained LLM) on the current buffer of problem instances and correct solutions $\mathcal{D}_{\text{GEN}}$. Generated solutions that yielded a correct answer are added to $\mathcal{D}_{\text{GEN}}$ to be used in future iterations, and all the generated solutions (correct and incorrect) are added to $\mathcal{D}_{\text{VER}}$. The verifier V^t is trained using DPO with a preference dataset constructed from pairs of correct and incorrect solutions from $\mathcal{D}_{\text{VER}}$. **Right:** At test time, the verifier is used to rank solutions produced by the generator. Such iterative training and inference-time ranking yields large improvements over generator-only self-improvement.

V-STaR



(a) GSM8K → MATH Subset



(b) MBPP → HumanEval

Order is important?

Premise Order Matters in Reasoning with Large Language Models

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Order is important?

✓ Sorted → correct

Rules: If duve, then dax. If kere and riff, then tess. If feg and tess, then rabe. If wozz, then riff. If rabe, then thoxe. If ag and riff and kere, then hepe. If thoxe, then vepe. If nife and hume and ag, then dax. If feg and thoxe, then ire. If rebe and ene and heff, then kone. If ire and dax, then wope. If tess and vepe, then nup. If rabe and vide and nife, then ag. If nup and ag and vide, then duve. If zeck and hepe and dax, then riff. If nup, then hepe. If sene, then hume. If hepe and tess and ag, then kere. If fum, then vide. If kere and wope, then fum. If jag and kone, then thoxe. If fum, then wozz.

Facts: Alice is cote. Alice is kone. Alice is duve. Alice is ag. Alice is jag. Alice is tess. Alice is riff. Alice is feg. Alice is vide.

Query: Is Alice wozz? In each step, indicate whether you are using a fact, or a rule.



... since Alice is fum, by the rule "If fum, then wozz," Alice is wozz. Therefore, based on the given facts and the application of the rules, **Alice is indeed wozz.**

✗ Shuffled → incorrect

Rules: If nup, then hepe. If kere and riff, then tess. If feg and tess, then rabe. If wozz, then riff. If tess and vepe, then nup. If ag and riff and kere, then hepe. If feg and thoxe, then ire. If nife and hume and ag, then dax. If ire and dax, then wope. If rebe and ene and heff, then kone. If hepe and tess and ag, then kere. If rabe, then thoxe. If rabe and vide and nife, then ag. If fum, then wozz. If zeck and hepe and dax, then riff. If kere and wope, then fum. If sene, then hume. If thoxe, then vepe. If fum, then vide. If duve, then dax. If jag and kone, then thoxe. If nup and ag and vide, then duve.

Facts: Alice is cote. Alice is kone. Alice is duve. Alice is ag. Alice is jag. Alice is tess. Alice is riff. Alice is feg. Alice is vide.

Query: Is Alice wozz? In each step, indicate whether you are using a fact, or a rule.



... we haven't established a clear path to "fum" based on the given facts and rules... based on the provided facts and the logical steps followed, **we cannot conclude that Alice is wozz.**

Order is important?

✓ Original → correct

Thomas withdraws \$1000 in 20 dollar bills from the bank account. **He loses 10 bills while getting home.** After that, he uses half of the remaining bills to pay for a bill. Thomas then triples his money. He then converts all his bills to 5 dollar bills. How many 5 dollar bills does he have?

Let's break down the problem step by step:

1. Thomas withdraws \$1000 in \$20 bills.
 $\$1000 / \$20 = 50$ bills
 2. **He loses 10 of those bills. $50 \text{ bills} - 10 \text{ bills} = 40 \text{ bills}$**
 3. He uses half of the remaining bills to pay for a bill. Half of 40 bills is 20 bills. $40 \text{ bills} - 20 \text{ bills} = 20 \text{ bills}$ remaining
 4. Thomas then triples his money. He has 20 bills of \$20, which is $20 * \$20 = \400 . Tripling \$400 gives us $\$400 * 3 = \1200 .
 5. He converts all his money to \$5 bills.
 $\$1200 / \$5 = 240$ bills
- So, Thomas has 240 five-dollar bills.



✗ Reordered → incorrect

Thomas withdraws \$1000 in 20 dollar bills from the bank account. After getting home, he uses half of the remaining bills to pay for a bill. Thomas then triples his money. He then converts all his bills to 5 dollar bills. **He lost 10 bills while getting home.** How many 5 dollar bills does he have?

Let's break down the problem step by step:

1. Thomas withdraws \$1000 in \$20 bills from the bank. $\$1000 / \$20 = 50$ bills
 2. He uses half of the remaining bills to pay for a bill. Half of 50 bills is 25 bills. So, he uses 25 bills and has 25 bills left.
 3. Thomas then triples his money. Since he had 25 bills of \$20 each, he had $25 * \$20 = \500 before tripling his money. Tripling \$500 gives him $\$500 * 3 = \1500 .
 4. He then converts all his bills to \$5 bills.
 $\$1500 / \$5 = 300$ bills
 5. **He lost 10 bills while getting home. $300 \text{ bills} - 10 \text{ bills} = 290 \text{ bills}$**
- So, Thomas has 290 \$5 bills.



Logical Reasoning w/ Distracting Rules

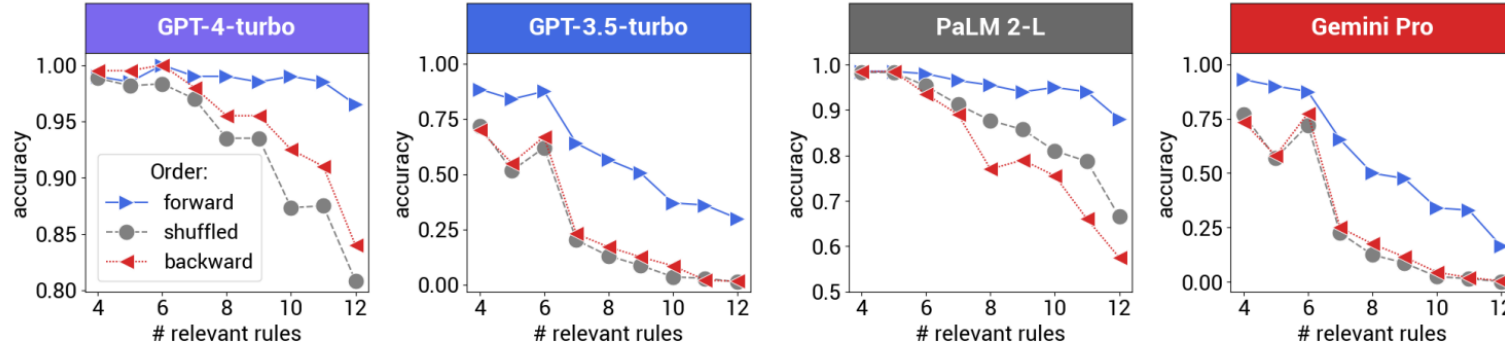
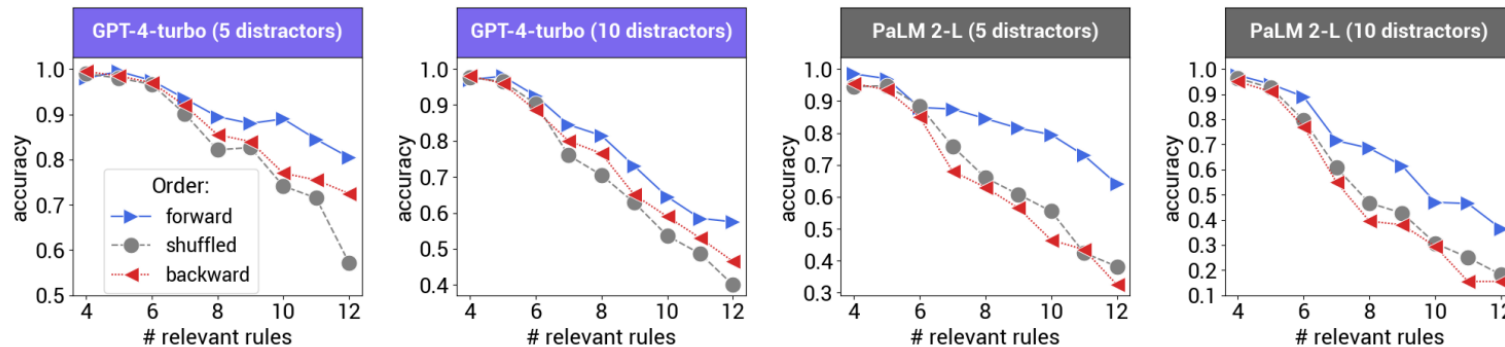


Figure 3 | Logical reasoning without distracting rules. See Table 6 in Appendix E for accuracy numbers.



- LLMs **fail to generate the proof** after changing the order of relevant rules.

while all LLMs perform best when the premise order follows the ground truth proof

Logical Reasoning w/ Distracting Rules

- Presenting “If A then B” before “If B then C” in the prompt generally achieves a **higher** accuracy compared to the reversed order.
- The performance gap is more significant when the number of premises **increases**.
- Even in reasoning domains where the **premise order does not matter**, premise order **does matter** in LLM reasoning.

Multi-agent Debate

Improving Factuality and Reasoning in Language Models through Multiagent Debate

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Multiagent Debate Improves Reasoning and Factual Accuracy

- Inspired by *The Society of Mind* and multi-agent settings.
- Multiple LLMs:
 - individually propose and jointly debate their responses and reasoning processes
 - to arrive at a **single common answer**.

Multi-agent Debate Improves Reasoning and Factual Accuracy

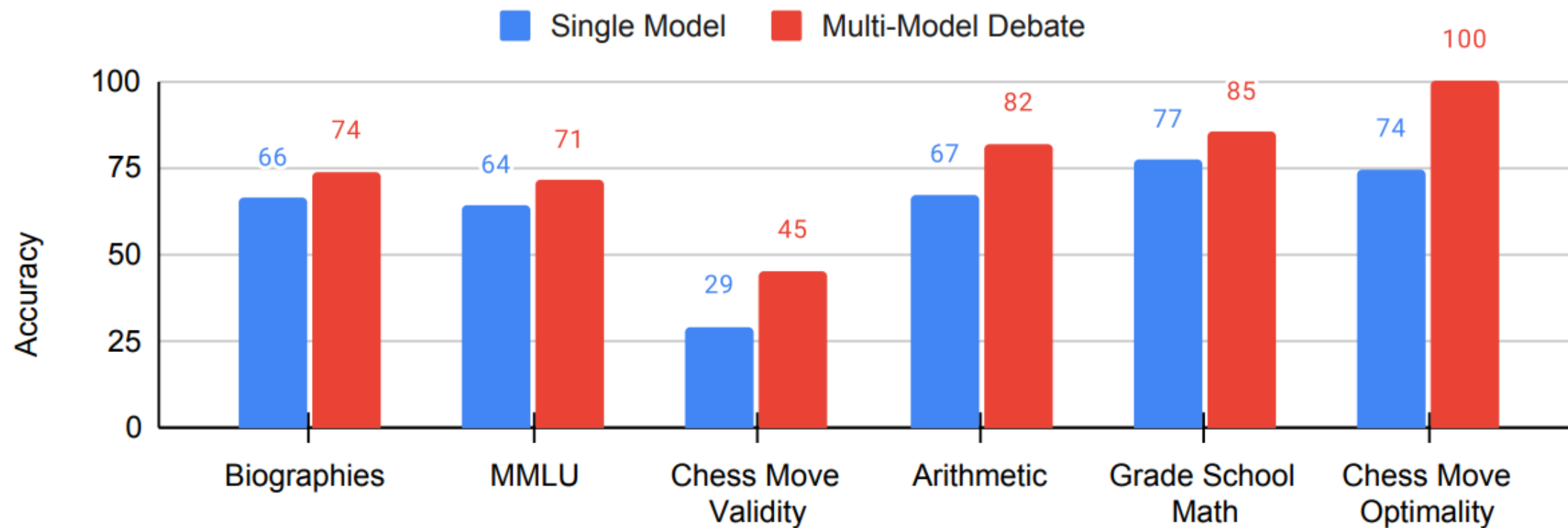


Figure 1: **Multiagent Debate Improves Reasoning and Factual Accuracy.** Accuracy of traditional inference and our multi-agent debate over six benchmarks (chess move optimality reported as a normalized score)

Example of Debate

User Input: A treasure hunter found a buried treasure chest filled with gems. There were 175 diamonds, 35 fewer rubies than diamonds, and twice the number of emeralds than the rubies. How many of the gems were there in the chest?

Round 1	
Agent 1: Let x be the number of rubies. Then the number of diamonds is 175. And the number of emeralds is $2(x-35)$, since there are twice as many emeralds as rubies and there are 35 fewer rubies than diamonds. To find the total number of gems, we add the number of diamonds, rubies, and emeralds: $\text{Total gems} = 175 + x + 2(x-35) = 3x + 105$. We don't know the exact value of x, but we do know that it represents the number of rubies, which is less than 175 (since there are 35 fewer rubies than diamonds). Let's try some values of x: If $x = 50$, then the total gems $= 3(50) + 105 = 255$. If $x = 100$, then the total gems $= 3(100) + 105 = 405$. Since we know there were fewer rubies than diamonds, the answer must be closer to 255 than to 405. After some trial and error, we can find that $x = 40$ gives us: $\text{Total gems} = 3(40) + 105 = 225$.	Agent 2: If there were 175 diamonds, then the number of rubies is 35 fewer, or $175 - 35 = 140$ rubies. And since there were twice as many emeralds as rubies, we can represent the number of emeralds as $2 * 140 = 280$ emeralds. To find the total number of gems in the chest, we can add up the number of diamonds, rubies, and emeralds: $175 + 140 + 280 = 595$. Therefore, there were 595 gems in the chest.
User Input: Using the solutions from other agents as additional information, can you give an updated response.... (Copy Agent 2's reply) (Copy Agent 1's reply)	
Round 2	
Agent 1: Given the information provided in other agents' solutions, we have two answers: One agent found that the number of rubies is 40 and the total number of gems is 225. Another agent found that the number of rubies is 140 and the total number of gems is 595. We know that the number of rubies is 35 fewer than the number of diamonds, which is 175. So the number of rubies should be $175 - 35 = 140$, as the second agent found. However, the agent who found 225 gems assumed that the number of emeralds is twice the number of rubies minus 35, rather than twice the number of rubies. If we use this assumption, we get: Number of rubies = 140. Number of emeralds $= 2 * 140 - 35 = 245$. Total number of gems $= 175 + 140 + 245 = 560$. Our final answer is 560.	Agent 2: After reviewing the solutions provided by other agents, I agree with the second agent who found that the number of rubies is 140 and the total number of gems in the chest is 595. Therefore, my answer to the math problem is 595.
User Input: Using the solutions from other agents as additional information, can you give an updated response.... (Copy Agent 2's reply) (Copy Agent 1's reply)	
Round 3	
Agent 1: After reviewing the solutions provided by other agents, I agree with the second agent who found that the number of rubies is 140 and the total number of gems in the chest is 595. Therefore, my answer to the math problem is 595.	Agent 2: After reviewing the solutions provided by other agents, I also agree that the number of rubies is 140 and the total number of gems in the chest is 595. Therefore, my answer to the math problem is also 595.

Figure 2: **Illustration of Debate.** Illustration of the debate procedure.

Multi-agent Debate

- population almost **always** converges on a single and more accurate common answer.
- Debate results are also less likely to include false facts.
 - as the debate progresses, individual model instances tend to disagree on uncertain facts and omit them from the answer

Example

- Debate for generating biographies of computer scientists

Prompt: Give a bullet point biography of David S. Johnson highlighting their contributions and achievements as a computer scientist, with each fact separated with a new line character.

Round 1

Agent 1: -- He earned his bachelor's degree from **Dartmouth College** in 1967 and his Ph.D. from MIT in 1973. ❌

-- Johnson has received numerous awards for his research and contributions to computer science, including the **ACM Turing Award** in 2013. ❌

-- Johnson's research interests include optimization, algorithms, and complexity theory. ✔️

Agent 2: -- David S. Johnson is an American computer scientist and mathematician. ✔️

-- Johnson was elected to the National Academy of Engineering in 2006. ✔️

-- He has been awarded numerous honors and awards for his contributions to computer science, including the **ACM SIGMOD Innovations Award** and the Knuth Prize. ❌

Round 2

Agent 1: -- David S. Johnson is a computer scientist and mathematician. ✔️

-- Johnson was elected to the National Academy of Engineering in 2006. ✔️

-- He has received numerous awards for his contributions to computer science, including the **ACM Turing Award** in 2013 and the Knuth Prize. ❌

Agent 2: -- David S. Johnson is an American computer scientist and mathematician. ✔️

-- He has worked at several prestigious universities, including Columbia University, AT&T Bell Laboratories, and **Cornell University**. ❌

-- Johnson is known for his work in the field of algorithms, particularly in the areas of network optimization, combinatorial optimization, and scheduling theory. ✔️

- All the models initially make incorrect predictions,
- but then arrive at the correct answer as debate progresses

Question: Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Round 1

ChatGPT: Before the restart, Carla was able to download 40% of the file, which is $0.4 * 200 \text{ GB} = 80 \text{ GB}$. At a rate of 2 GB/minute, it would take her: $\text{time} = 80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes}$.

Because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the remaining 120 GB is: $\text{time} = 120 \text{ GB} \div 2 \text{ GB/minute} = 60 \text{ minutes}$.

But Carla also had to wait 20 minutes for the Windows update to install, so the total time it took to download the file is: $\text{total time} = 40 \text{ minutes} + 60 \text{ minutes} + 20 \text{ minutes} = 120 \text{ minutes}$. ❌

Bard: Carla is downloading a 200 GB file at a speed of 2 GB/minute. This means that it will take her $200 / 2 = 100 \text{ minutes}$ to download the file.

However, Windows forces a restart to install updates, which takes 20 minutes. This means that Carla will have to restart the download from the beginning. Since Carla has already downloaded 40% of the file, she only needs to download $200 - 80 = 120 \text{ GB}$. At a speed of 2 GB/minute, it will take Carla $120 / 2 = 60 \text{ minutes}$ to download the file.

Adding the time for the Windows restart, it will take Carla $80 + 20 = 100 \text{ minutes}$ to download the file. ❌

Round 2

ChatGPT : Based on the calculations provided by the other agent, the correct answer is: Before the restart, Carla was able to download 40% of the file, which is $0.4 * 200 \text{ GB} = 80 \text{ GB}$. At a rate of 2 GB/minute, it would take her: $\text{time} = 80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes}$

However, because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the entire 200 GB file is: $\text{time} = 200 \text{ GB} \div 2 \text{ GB/minute} = 100 \text{ minutes}$

And since Carla had to wait 20 minutes for the Windows update to install, the total time it took to download the file is: $\text{total time} = 40 \text{ minutes} + 100 \text{ minutes} + 20 \text{ minutes} = 160 \text{ minutes}$. ✅

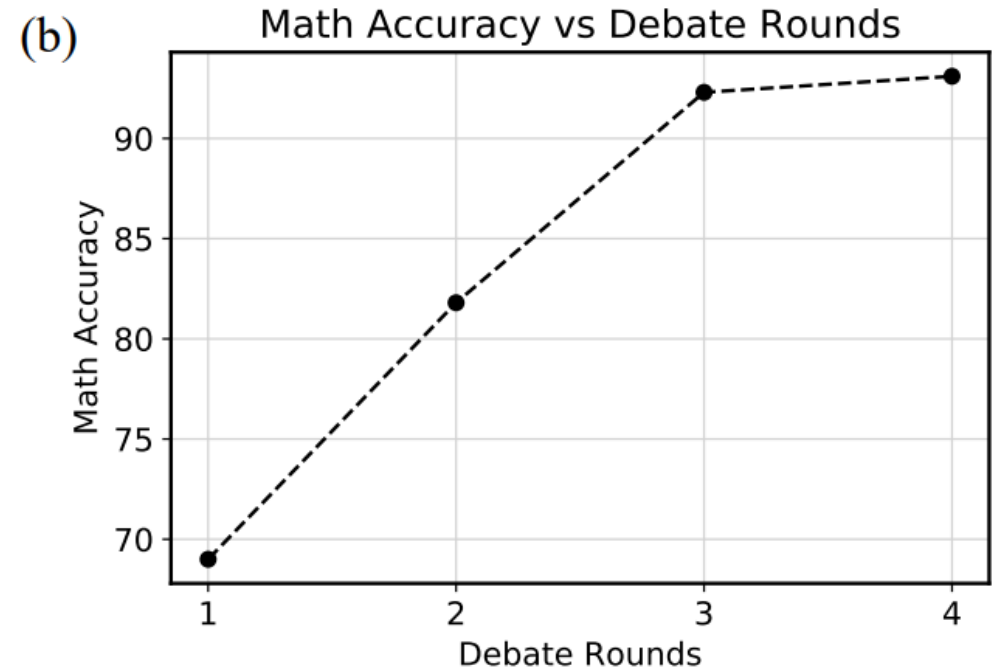
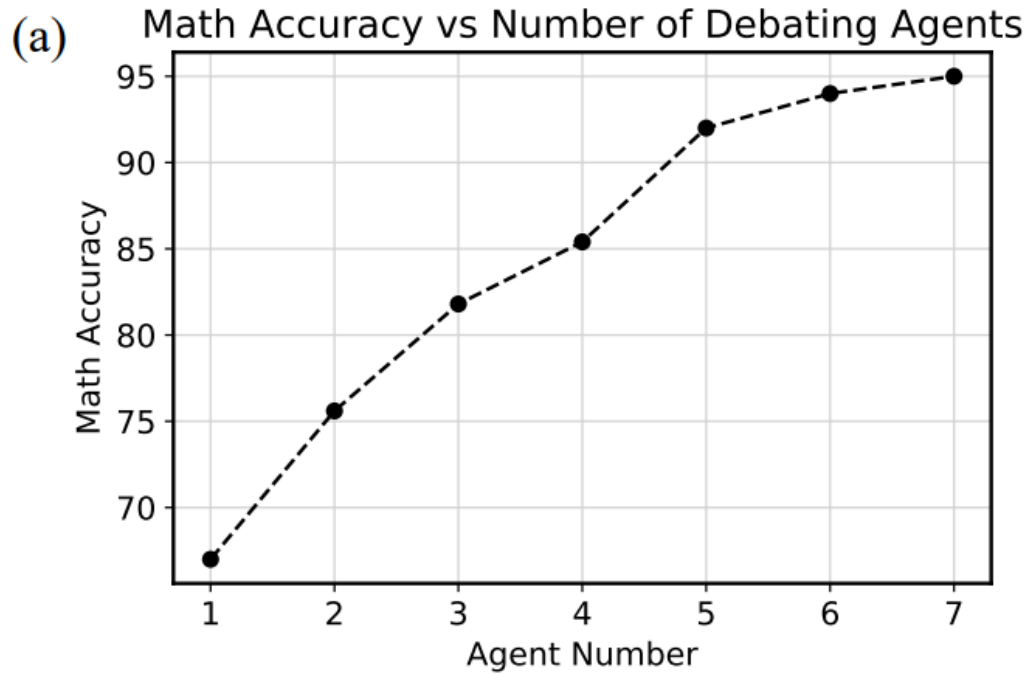
Figure 11: **Debate Between chatGPT and Bard** Illustration of debate between different models.

Example for Solving Math

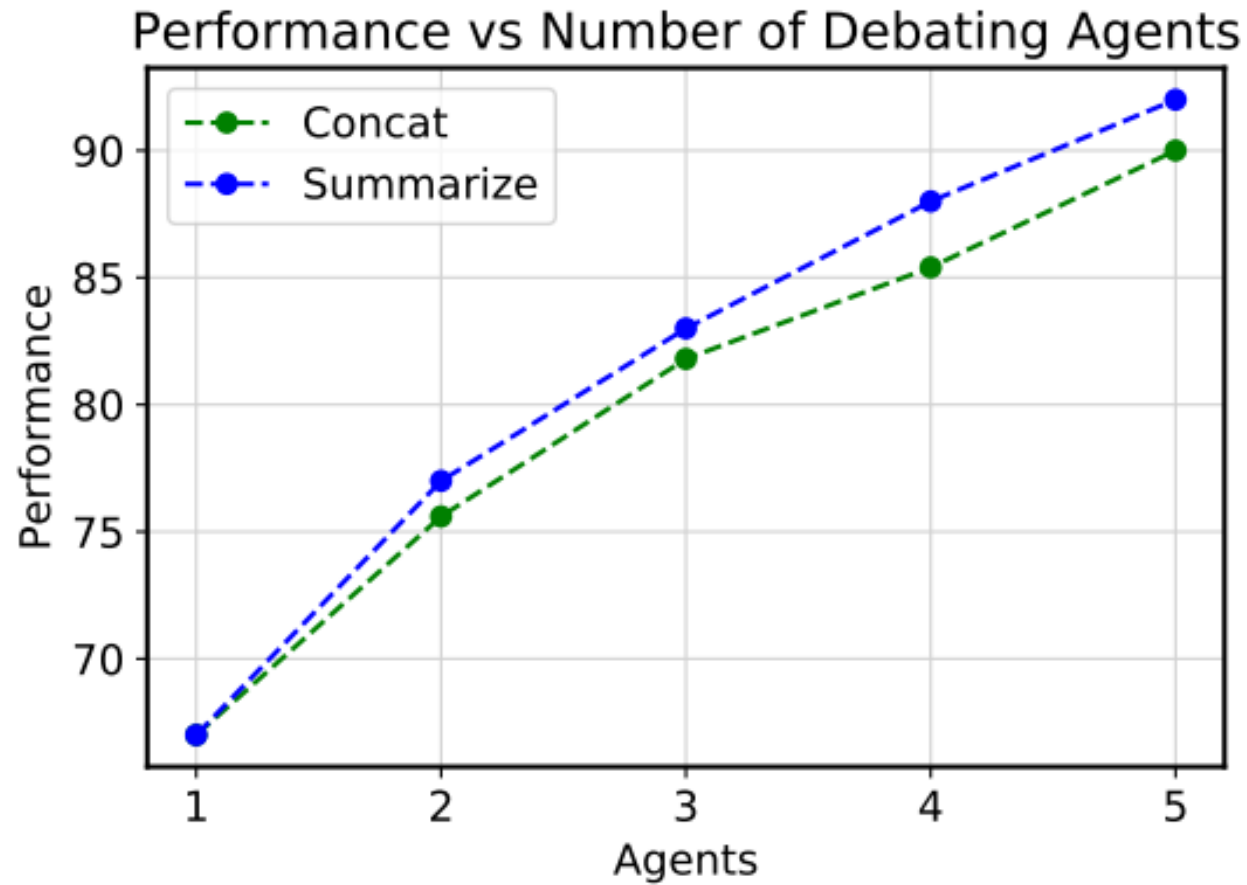
	Question: What is the result of $10+20*23+3-11*18$?		Question: What is the result of $3+7*9+19-21*18$?		
Round 1	Agent 1: 269 ❌	Agent 2: 369 ❌	Agent 1: 378 ❌	Agent 2: -351 ❌	Agent 3: -357 ❌
Round 2	Agent 1: 275 ✅	Agent 2: 275 ✅	Agent 1: -293 ✅	Agent 2: -293 ✅	Agent 3: 19 ❌
	Question: What is the result of $4+23*6+24-24*12$?		Question: What is the result of $8+14*15+20-3*26$?		
Round 1	Agent 1: -244 ❌	Agent 2: -146 ❌	Agent 1: 236 ❌	Agent 2: -214 ❌	Agent 3: 210 ❌
Round 2	Agent 1: -146 ❌	Agent 2: -122 ✅	Agent 1: 160 ✅	Agent 2: 160 ✅	Agent 3: 160 ✅
Round 3	Agent 1: -122 ✅	Agent 2: -122 ✅	Agent 1: 160 ✅	Agent 2: 160 ✅	Agent 3: 160 ✅

Figure 4: **Illustration of Solving Math.** Reasoning between agents is omitted.

Performance with Increased Agents/Rounds



Effect of Summarization

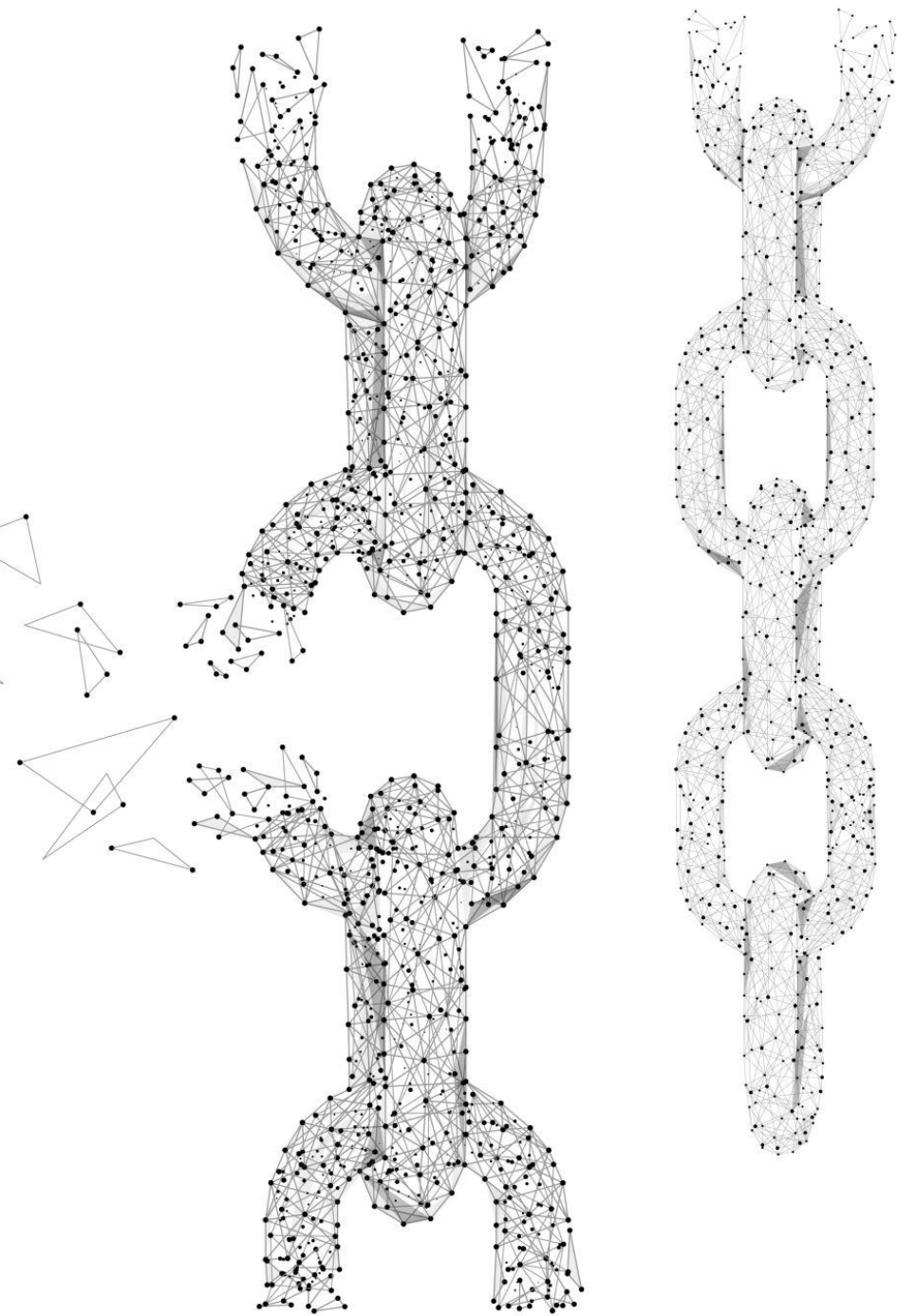


Chain of Code (CoC)

Chain of Code: Reasoning with a Language Model-Augmented Code Emulator

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Li Fei-Fei¹ Fei Xia^{* 2} Brian Ichter^{* 2}

<https://chain-of-code.github.io/>



Code
is
advantageous

Direct answer only

Q: How many countries have I been to? I've been to Mumbai, London, Washington, Grand Canyon, ...

A: 32 (20%, ✗), 29 (10%, ✗), 52 (10%, ✓), ...

Chain of Thought

Q: Let's think step by step. How many countries have I been to? I've been to Mumbai, London, ...

We'll group by countries and count:

1. India: Mumbai, Delhi, Agra
2. UK: London, Dover, Edinburgh, Skye
3. USA: Washington, Grand Canyon, ...

A: 61 (20%, ✗), 60 (20%, ✗), 52 (10%, ✓), ...

Chain of Code (Ours)

Q: How many countries have I been to? I've been to Mumbai, London, Washington, Grand Canyon, Baltimore, ...

```
1  places, countries = ["Mumbai", ...], set()
   delta state:  {places = ['Mumbai', ...], countries =
   set()}

2  for place in places:
   delta state:  {place = 'Mumbai'}

3  country = get_country(place)
   delta state:  {country = 'India'}}

4  countries.add(country)
   delta state:  {countries = {'India'}}

5  answer = len(countries)    delta state:  {answer =
   52}
```

A: 52 (100%, ✓)

Result

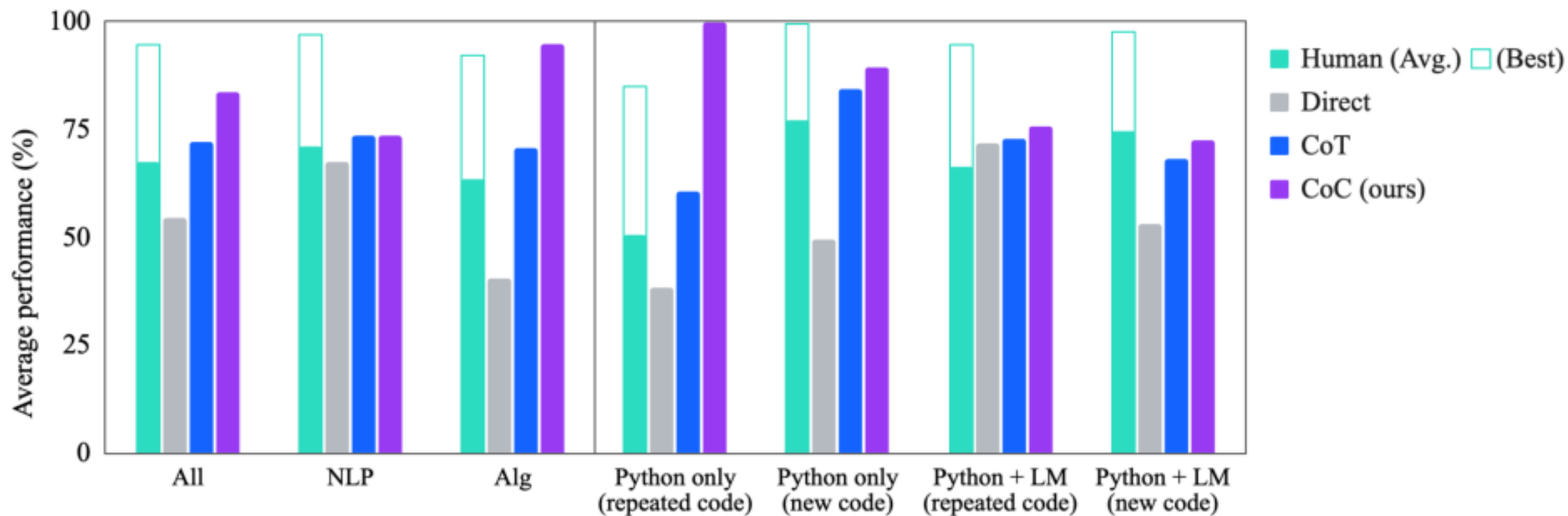


Figure 4. Average performance across different baselines grouped by task type, indicating the problem type and how CoC is generated & executed.

A Big Question!

- Many semantic tasks are rather difficult/impossible to express in code.
 - E.g. sarcasm in a string.